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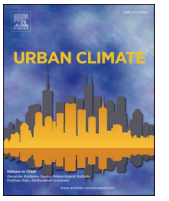
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City-scale evaluation of urban ecohydrologic processes on surface energy balances, heat, and humidity

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ABSTRACT

Urban regions substantially modify both surface energy and hydrologic cycles. Despite the linkage between urban hydrology and energy cycles, modern coupled land-atmosphere models do not represent common urban hydrologic features like runoff routing from impervious to pervious surfaces or tree canopy that shades pavements. We compare three urban surface models in the Weather and Research Forecasting model across multiple hydrometeorological events in Milwaukee, Wisconsin: a widely used slab urban model (Noah-MP Bulk Parameterization; *Typical*), multiple land covers per grid cell (Noah-MP Mosaic; *Mosaic*), and a model which adds sub-grid water transfers between land cover types (Noah-MP HUE; *HUE*). Inclusion of urban hydrology and vegetation (*HUE*) increases albedo and emissivity, reducing available energy in our study region. Ambient soil moisture conditions cause divergent responses in *HUE* simulations: warming when soil water is limited and cooling when ample soil water is available for evapotranspiration. Comparison against observed 2m air temperature and specific humidity show increased skill in the *HUE* model simulations, especially compared to the *Typical* model. Noah-MP HUE presents a stride in understanding how urban hydrology influences city-scale meteorology and a pathway to examine urban hydrologic greening initiatives more wholistically in regional atmospheric models.

1. Introduction

Cities continue to expand as the world's population grows (United Nations, 2018), exacerbating issues associated with urbanization. Urbanization replaces natural land cover with impervious engineered surfaces like buildings, roadways, and sidewalks. This process of replacing natural ecosystems with pavement-rich environments that contain darker, densely packed buildings causes more radiative energy to be stored (Christen and Vogt, 2004; Oke, 1982; Oke et al., 2017). The resulting replacement of vegetation reduces evaporative cooling (Aram et al., 2019; Bowler et al., 2010; Gunawardena et al., 2017; Knight et al., 2021) and leads to urban heat islands, where city temperatures are elevated relative to surrounding areas (Oke, 1982), impacting human comfort and health (Tong et al., 2021). Simultaneously, increasing the amount of paved surfaces enhances runoff generation and thus flood hazard due to the lack of infiltration capacity (Boyd et al., 1993; Hollis, 1975; Leopold, 1968; Yang et al., 2013). Due to these hazards and higher risks of physical and economic loss in cities (Hammond et al., 2015), urban spaces have become a particular focus of mitigation strategies.

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Many hydrologic mitigation strategies focus on nature-based greening initiatives that aim to reintroduce natural vegetation and increase infiltration (Aram et al., 2019). Green initiatives, which typically focus on adding nature-based solutions like green infrastructure or vegetation to reduce runoff also provide multiple mechanisms for lowering urban surface and air temperatures (Gunawardena et al., 2017; Oke, 1989; Sanusi et al., 2016). Surface temperatures can be greatly reduced by tree canopies due to shading (Du et al., 2021; He et al., 2024; Schwaab et al., 2021, among many others), though the extent of cooling depends on the spatial configuration of the trees and climate conditions (Jung et al., 2021; Morabito et al., 2021; Zhou et al., 2017). Air temperatures are also reduced through both shading and enhanced evaporative cooling from tree canopies and vegetation (Bowler et al., 2010; Gobatti et al., 2025; Knight et al., 2021). While both shading and evaporative cooling reduce surface and air temperatures, the effects at the city-scale can be difficult to quantify due to changes in day-to-day meteorology (Ettinger et al., 2024; Lachapelle et al., 2023; Vo and Hu, 2021; Wang et al., 2023a; Ziter et al., 2019). Nevertheless, nature-based greening initiatives have been shown to help mitigate the impacts of heatwaves that are enhanced by urbanization (Founda and Santamouris, 2017; Kong et al., 2021; Ramamurthy and Bou-Zeid, 2017; Schatz and Kucharik, 2015; Ward et al., 2016; Zhao et al., 2018),

Coupled land-atmosphere models are a promising tool to evaluate the potential impacts of urban greening initiatives on temperature and near surface humidity at the city-scale. For instance, the Weather Research and Forecasting (WRF) model has been used to quantify urban impacts on atmospheric circulation for various cities across the world at fine spatial scales (e.g. 1-km in Sharma et al., 2016b, 0.5-km in Papangelis et al., 2012, 0.33-km in Jandaghian and Berardi, 2020, etc.) and at temporal scales ranging from annual (De La Paz et al., 2016; Feng et al., 2015) to seasonal (Cui and de Foy, 2012; Yang et al., 2012), daily, and sub-daily (Giannaros et al., 2013; Li et al., 2018; Sharma et al., 2016b; Wang and Li, 2021). Within WRF and other atmospheric models, urban land surface representations—which range in complexity from simple bulk models that creates grid-cell-wide impervious surfaces with modified thermal and surface roughness (Liu et al., 2006) to models integrate the so-called urban canyon with single (Kusaka et al., 2001; Kusaka and Kimura, 2004; Yang et al., 2015) or multiple layers (Chen et al., 2011; Martilli et al., 2002a; Salamanca and Martilli, 2010)—prioritize modeling sensible and ground heat fluxes (Oke, 1982; Oke et al., 2017; Stull, 1988) due their effects on the evolution of atmospheric processes (Sharma et al., 2021).

Coupled land-atmosphere models currently do not include urban ecohydrologic processes that change evaporative cooling, which reduces model skill in these critical regions during high impact events. Specifically, limited urban hydrologic processes and minimal representation of urban vegetation may cause drier, hotter conditions than those observed (Wang et al., 2019). Further, this lack of hydrologic and vegetation process representation hampers efforts in assessing efficacy of possible urban greening initiatives at city-scale. Many urban greening initiatives include routing runoff from impervious to pervious areas using rain gardens or downspout disconnection initiatives (see section 3.6 in Liu et al. (2021) and Eckart et al. (2017) for overviews of practices involved with urban greening plans), changing urban hydrologic balances at different scales (Arjenaki et al., 2021; Shields and Tague, 2015; Voter and Loheide II, 2018). This extra available soil water may increase evapotranspiration, but coupled land-atmosphere models like WRF cannot represent this interaction. Despite its ubiquity in the real world and explicit inclusion in greening plans (Sousa-Silva et al., 2023), urban tree canopy that overhangs pavement, which shades underlying surfaces, increases evapotranspiration, and intercepts rainfall (Selbig et al., 2022) is not represented in any coupled land-atmosphere model to date. Instead, models sometimes address urban regions through integrating "mosaicking schemes" take the *N* most abundant land covers within a single model grid, simulates each land cover individually, and average the outputs based on their relative areas (Li et al., 2013). While mosaic schemes increase model skill, especially in urban regions (Jänicke et al., 2017; Li et al., 2018; Sharma et al., 2016b; Wang et al., 2022; Zhang et al., 2021), they do not integrate processes that 'overlap' or 'communicate' across land covers co-located within a larger atmospheric model grid that are inherently needed to quantify urban greening impacts. While some efforts have been made to add urban vegetation or hydrology to city-scale models, resulting parameterizations are often too computationally expensive and neglect hydrologic processes (Krayenhoff et al., 2020), contain no tree shading of pavement or runoff routing (Belda et al., 2021; Hollósi et al., 2024), focus exclusively on tree canopy radiative effects with no hydrologic process integration (Krayenhoff et al., 2014), or do not transfer water or explicit tree canopy shading via overhanging (Ryu et al., 2025).

In this study, we aim to investigate the importance of urban ecohydrology within coupled land-atmospheric simulations by using WRF and a recent urban land surface model called Noah-MP for Heterogenous Urban Environments (HUE; Alexander et al., 2024). Noah-MP HUE was specifically developed to represent the impacts of subgrid-scale lateral transfers of water and energy, and thus may lead to increases in latent heat (*LH*) and change surface meteorology at city-scale. Our study focuses on Milwaukee, Wisconsin, a mid-sized humid continental city in the upper midwestern United States along the west shore of Lake Michigan. We run simulations across different hydrometeorological conditions to investigate the impact of urban ecohydrology (e.g., tree canopy shading pavement and downspout disconnection) on city-scale surface energy balances, air temperature, and specific humidity. We test three simulation types, the default dominant land cover simulation which simulates a single land cover per grid cell (He et al., 2023; Niu et al., 2011), a land cover mosaicking scheme implemented into Noah-MP (Alexander et al., 2024; Li et al., 2013), and Noah-MP HUE, which includes the mosaicking scheme as well as certain lateral water transfers that are unique to urban environments. Specifically, we aim to answer:

1. How does the representation of urban heterogeneity and urban hydrology, in the form of lateral transfers of water and energy, influence available energy at the city-scale?
2. When lateral transfers of water and energy are represented in coupled simulations, what are the effects on partitioning of surface energy?
3. When lateral transfers of water and energy are represented in coupled simulations, is there a notable change in simulated daytime surface air temperature and humidity compared to other common model representations and in-situ observations?

2. Methods

2.1. Overview of numerical experiments and Noah-MP HUE

We employed three separate model formulations of Milwaukee (Fig. 1A). All formulations used Noah-MP, a modern land surface model that is used to interpret atmospheric forcings into the spatiotemporal evolution of water, energy, and carbon fluxes of the near surface (He et al., 2023; Niu et al., 2011).

The first model formulation uses the default urban option present within the WRF model. This is traditionally termed a 'bulk' approach, in which urban regions are parameterized only as impervious surfaces. There is no urban vegetation, and only properties like albedo, emissivity, and roughness lengths are changed based on the intensity of urban category (Chen et al., 2011; Liu et al., 2006). This model type (e.g. a slab representation of urban land cover) is used within many climate modeling and weather forecasting contexts (Lipson et al., 2024) including the recent high resolution CONUS404dataset generated by the National Center for Atmospheric Research and United States Geological Survey (Rasmussen et al., 2023). We adopted this bulk model for our simulations, referred to as '*Typical*' throughout the rest of this study. The choice to use this bulk model, rather than other more sophisticated urban representations available in Noah-MP, is described further below.

The second model formulation uses multiple land cover types per grid square, referred to as a mosaicking scheme (Alexander et al., 2024; Li et al., 2013). Both the dominant land cover type in *Typical* and the next N land cover types, $N = 8$in this study, are simulated by Noah-MP. Then, these land cover types are averaged together using weights based the relative fraction of the land cover and output or passed onto other aspects of the atmospheric model (Fig. 1C). Our model passes heat fluxes (H, LH, G), emergent parameters like

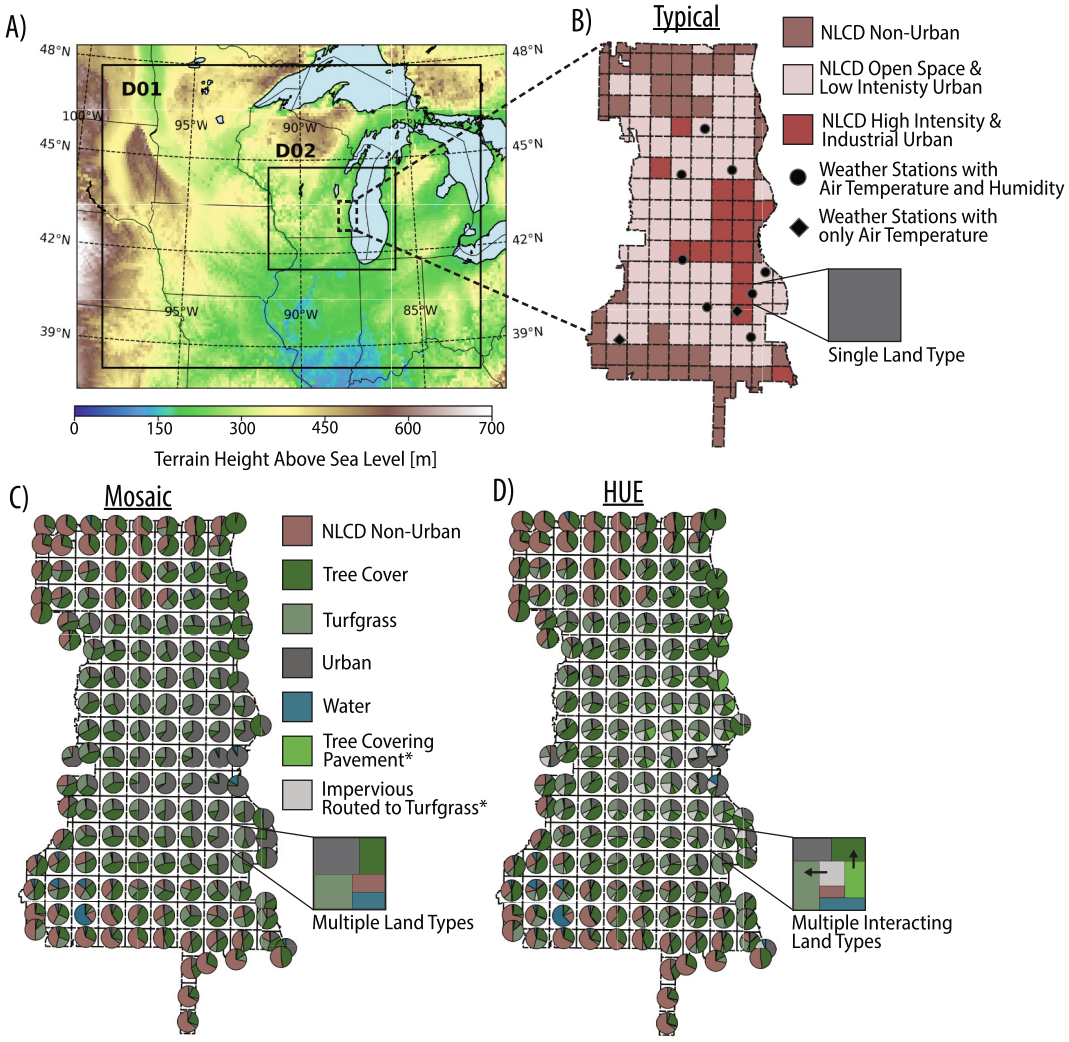


Fig. 1. WRF terrain map over the 9-km (D01) and 3-km (D02) domains (A). Land cover maps of land surface distribution for *Typical* (B), *Mosaic*, (C) and *HUE* (D) simulations. Land covers demarcated with an asterisk are only active in the study region within HUE simulations. Weather stations that are used for comparison of simulated temperature and specific humidity are shown in the *Typical* land cover map (B).

effective roughness lengths and albedos, and surface meteorology like humidity and temperature to different modules of the WRF model. This formulation is referred to as '*Mosaic*' in this study.

The third model formulation uses the newly developed version of Noah-MP designed for Heterogeneous Urban Environments (*HUE*; Alexander et al., 2024). *Noah-MP HUE* integrates both urban heterogeneity (e.g. a mosaicking scheme) with hydrologic processes that pass water from one land cover to another. In this study, runoff is passed from impervious surfaces (e.g. pavements) to turfgrasses (downspout disconnection) as described in Alexander et al. (2024). We adopt the same parameters for turfgrass as outlined in Alexander et al. (2024).The *HUE* formulation also models tree canopy that overhangs pavement: tree foliage shades an underlying pavement and transpires using soil moisture extracted from intra-grid tiles representing trees (Alexander et al., 2024). As many urban greening plans aim to mitigate urban heat impacts through enhanced evapotranspiration and shading of pavements through tree cover expansion and routing of runoff, this formulation aims to test the potential of these processes in coupled atmospheric modeling at the city-scale under current land cover conditions in order to provide a baseline for future management-based scenarios.

Importantly, we did not incorporate any of the more detailed urban parameterizations that are available in the WRF model in any of our three model formulations. Parameterizations like the single layer urban canyon model (Kusaka et al., 2001; Kusaka and Kimura, 2004; Yang et al., 2015) and BEP + BEM scheme (Chen et al., 2011; Martilli et al., 2002; Salamanca and Martilli, 2010) are widely used in investigations of urban meteorology. These parameterizations offer more detail in urban areas, particularly for radiative transfer and turbulent exchanges from different urban components. Unfortunately for this study, this means that there are structural differences in how parameters are used in these urban models relative to how they are used in Noah-MP, which handles calculations for all other land cover classes including our novel land covers, such as tree canopy that overlaps pavement. Using those more sophisticated urban models in this study would thus leave us unable to untangle to what extent changes in outputs were driven by ecohydrologic processes vs. by structural differences and model averaging. Furthermore, because the single layer urban canyon model and BEP + BEM model do not contain any vegetation or related surface hydrology, we could not integrate ecohydrologic processes into the detailed urban models without significant effort. Therefore, to ensure all changes in outputs are due exclusively to the effects of ecohydrologic processes, we utilized the bulk urban scheme for all WRF simulations, including any pavements in novel land cover classes in the *Mosaic* and *HUE* simulations.

Land cover maps are presented in Fig. 1for *Typical* (Fig. 1B), *Mosaic* (Fig. 1C), and *HUE* (Fig. 1D). Land cover types used in *Mosaic* and *HUE* were derived from high resolution imagery and GIS data (Section 2.2).

2.2. Derivation of custom land cover maps for Milwaukee

We used the 2011National Land Cover Database (NLCD) dataset to drive *Typical* simulations (Homer et al., 2015). Due to lack of information on fine scale urban features that are required for *HUE* simulations in widely used land cover datasets, such as urban tree covering impervious surfaces and impervious surfaces that are readily routed to pervious land covers, we derived custom land cover distributions using GIS and imagery data. To quantify tree cover, we classified imagery from the United States Geologic Survey National Agricultural Imagery Project (NAIP) over the Milwaukee Metropolitan Sewerage District (MMSD) planning region into 'tree' or 'not a tree' based polygons (Earth Resources Observation And Science (EROS) Center, 2017). Characteristics of colour, texture, and entropy extracted from raw NAIP imagery provided the foundation of the ensemble of pixel level classifiers. These classifiers were used to weight a final pixel classification of 'tree' (Bosch, 2020; Hastie et al., 2009; Yang et al., 2009). 1 % of the MMSD region, labelled by hand, served as training data for classification.

Urban impervious surfaces were drawn primarily from LiDAR-based GIS layers from both 2015and 2020 to cover the entire MMSD planning region. These LiDAR layers of impervious surfaces contained information on roadways, building footprints, and sidewalks, which were then modified to contain only one type of impervious surface per layer. We found that highway interchanges were underrepresented in the roadway layers and many sidewalks were missing in suburban regions adjacent to major roadways (Light Pink pixels in in Fig. 1B) compared to imagery data. To address these discrepancies, roadway widths over highway overpasses were increased by 400 % and roadway widths in suburban areas were increased by 7 %. The choice of a 400 % roadway width increase reduced the amount of missed urban impervious surfaces in the downtown region, shown in Fig. S1, while 7 % was based on manual classification in suburban areas. Finally, walkways leading up to houses and driveways were missing in many locations throughout MMSD. We accounted for these missing features by buffering the average width of existing sidewalks by 0.2m and buffering existing driveways by 0.5m to account for the single- or two-family tax parcels in the MMSD planning region that did not include front walkways or driveways in GIS layers.

These new surface data layers were then combined to create distributions of land cover within the MMSD region which we applied within the inner, 3km WRF numerical grid of *Noah-MP HUE*. We determined the amount of area requiring classification by overlapping each pixel within the 3km WRF grid with the MMSD service area. Then, the following scheme was employed:

1. Waterbodies from the 24K Waterbody database hosted by the Wisconsin Department of Natural Resources were set to 'Water' (blue in Fig. 1D).
2. Tree cover that overlapped any impervious surface (buildings, buffered roadways, sidewalks, or driveways) were set to "Tree Covering Pavement" (bright green in Fig. 1D).
3. All other tree cover was set to "Tree" (dark green in Fig. 1D).
4. Remaining building footprints that were not covered by tree were classified as "Impervious Routing to Turfgrass" (bright grey in Fig. 1D).
5. All other impervious surfaces were categorized as "Urban" (dark grey in Fig. 1D).

6. Then, we add in NLCD based information for classification of grasslands and croplands (a majority of the peri-urban spaces in MMSD), which are presented as "NLCD Non-Urban" (red in Fig. 1D).
7. The remaining area in the MMSD region is then categorized as Turfgrass (sage green in Fig. 1D).

For *Mosaic* simulations, a similar classification scheme was used, but with two key differences: 1) There is no "Tree Covering Pavement". Instead, this tree canopy is classified as "Urban". 2) There is no "Impervious Routing to Turfgrass". Instead, all remaining building footprints were similarly classified as "Urban". Thus, the main differences between the *Mosaic* and *HUE* simulations are the latter's representation of pavement covered by trees (forest green to bright green in Fig. 1D), and runoff routed from urban spaces (proportional to the amount of urban space that is rooftop in a grid cell) to turfgrass (light grey to sage green in Fig. 1D).

2.3. Common WRF model specifications

Atmospheric response to urban hydrology and vegetation was simulated using the WRF model version 4.4.1 (Skamarock et al., 2019). We used a hindcast configuration of the WRF model with two one-way nested domains of 9and 3km horizontal resolutions and 51 terrain-following vertical levels. Simulations used ERA5 reanalysis as boundary conditions (Hersbach et al., 2020) and a common suite of physics options, including the RRMTG longwave and shortwave radiation schemes (Iacono et al., 2008), the Morrison Double Moment Cloud physics scheme (Morrison et al., 2009), the YSU planetary boundary layer scheme (Hong et al., 2006), and the revised MM5 surface layer scheme (Jiménez et al., 2012). Noah-MP specific options are provided in Table S1. All simulations update the Great Lakes sea surface temperatures every hour based on downscaled ERA5 reanalysis (Hersbach et al., 2020). Domain 1 employs the Kain-Fritsch cumulus parameterization scheme (Kain, 2004) and Domain 2uses no cumulus parameterization, as 3-km horizontal resolution is adequate to resolve scales associated with deep convection (Lucas-Picher et al., 2021). Simulation output for Domain 2occurred at 30-min time steps.

Simulation periods were selected based on differing hydrometeorological conditions, listed in Table 1. Ten simulations focused on rainfall events (number of days = 32), and two on longer-duration heatwaves (number of days = 19). The first 24h of each simulation were discarded as model spin-up (e.g. analysis started 24h after those listed in Table 1). Uncoupled "offline" continuous land surface simulations spanning from 2011 to 2021 of each land cover scenario (e.g. *Typical*, *Mosaic*, *HUE*) provided initial land-surface soil moisture and temperature profiles at the start of WRF simulations. These long term simulations were forced by two meter air temperature and precipitation from the Analysis of Record for Calibration (AORC; Fall et al., 2023) with surface pressure, ten meter winds, two meter humidity, and incoming shortwave and longwave radiation from surface ERA5 reanalysis (Hersbach et al., 2020). This combined AORC/ERA5 forcing dataset was gridded at a 4-km spatial resolution and hourly temporal resolution following Cosgrove et al. (2003), and was further re-gridded to match our WRF domain resolutions using a bi-linear approach that conserves the integral of the source resolution. More details can be found in (Abbasian et al., 2025) Finally, to minimize differences in WRF simulation results due to diverging large-scale meteorological conditions in Domain 1, each simulation of Domain 2is forced with the *Typical* simulation of Domain 1 (Skamarock et al., 2019).

2.4. Observations and model evaluation metrics

To evaluate the skill of each model formulation, we compared hourly 2 m air temperature (*T2*) and 2 m specific humidity (*Q2*) to observations within the MMSD planning region. Observation locations are shown in Fig. 1B. Seven locations measure both *T2* and *Q2*, and two more measure only *T2*. One station is hosted by the Automated Surface Weather Observing System at the Milwaukee Airport (KMKE), while the others are part of the Meteorological Assimilation Data Ingest System Network (accessed through the MesoWest interface hosted by the University of Utah; Horel et al., 2002a, 2002b; United States, 1998). *Q2* is not directly measured by our observations, so we use dewpoint temperature and surface pressure measurements to calculate specific humidity. Further, if an observation was not available due to internal quality controls from either of the networks we draw from, that time was dropped from analysis. Only $n = 320$ (2 % of total) observations were dropped in this manner for *T2* and a further $n = 143$ (1 % of total) were dropped

Table 1
WRF simulation abbreviated name, start date, end date, and hydrometeorological condition.

WRF Simulation Name	Simulation Start Date	Simulation End Date	Hydrometeorological Condition
July 2012	28June 201200 UTC	11 July 2012 00 UTC	Heatwave
June 2014	15 June 2014 00 UTC	19June 2014 00 UTC	Rainstorm
July 2015	4 July 2015 00 UTC	8 July 2015 00 UTC	Rainstorm
July 2016	19July 2016 00 UTC	27 July 2016 00 UTC	Heatwave
Sep 2016	5 September 2016 00 UTC	8 September 2016 12 UTC	Rainstorm
July 2017	9July 2017 00 UTC	13July 2017 00 UTC	Rainstorm
June 2018	16 June 201800 UTC	21 June 201800 UTC	Rainstorm
Sep 2018	1 September 2018 00 UTC	6 September 2018 12 UTC	Rainstorm
July 2019	18 July 2019 00 UTC	21 July 2019 12 UTC	Rainstorm
Sep 2019	8 September 2019 12 UTC	13 September 2019 00 UTC	Rainstorm
July 2020	6 July 2020 00 UTC	11 July 2020 00 UTC	Rainstorm
Aug 2021	5 August 2021 12 UTC	9 August 2021 00 UTC	Rainstorm

for *Q2*. We compared simulations to observed air temperatures and specific humidity using Mean Bias (MB), Root Mean Square Error (RMSE), and Pearson's Correlation Coefficient (Pearson's R). Negative and positive values of MB indicate underestimation and overestimation compared to observations, respectively. Note that *T2* observations were only reported to the nearest degree Fahrenheit. For a fair comparison, all observations and model simulations are first changed to degrees Fahrenheit, rounded to the nearest degree, and then changed back to degrees Celsius for comparison.

Unless otherwise specified, when rain occurs in simulations (*Typical*, *Mosaic*, or *HUE*), that time step was removed from the analysis. This ensured that the same sample size is used for all analyses, and compares similar conditions, as the timing of rainfall is not consistent between the simulations. Analysis of rainfall for these simulations is the topic of a separate ongoing study.

3. Results

3.1. Changes to urban radiative properties lead to differences in available energy

Urban heterogeneity, whether introduced through the combination of more realistic urban hydrologic processes paired with multiple land cover types (*HUE*) or just the latter (*Mosaic*), leads to higher broadband surface albedo (α) and emissivity (ε) in simulations throughout MMSD planning region (Fig. 2). This change is proportional to the amount of new vegetation that is represented in simulations. Urban regions in *Typical* simulations use a fixed α of 0.113 and ε of 0.91 (Fig. 2 A and 2F), while both *Mosaic* and *HUE* simulations produce dynamic variable values that are generally higher and more spatially variable, especially in suburban regions for both α (*Mosaic* average: 0.155, *HUE* average: 0.156) and ε (*Mosaic* average: 0.991, *HUE* average: 0.992). The differences between *Mosaic* and *HUE* are subtle, with *HUE* producing small increases in α (+0.001) and ε (+0.002) on average. The only difference driving increases in α and ε is the introduction of the tree over pavement land cover (bright green in Fig. 1D). Locations that represent large amounts of tree cover overlapping pavement in *HUE* generate larger differences in α and ε when compared to *Mosaic*, and these differences are particularly large during times of high solar zenith angles, due to the diurnal evolution of vegetated α and ε within Noah-MP (He et al., 2023). Differences between *Mosaic* and *HUE* are marginal compared the inclusion of vegetation itself (e.g. using *Mosaic* or *HUE* realizations instead of the *Typical* model).

Changes in α and ε lead to decreases in net radiation in predominantly urban areas and increases in non-urban areas within the MMSD planning region (Fig. 3). In non-urban areas, there are modest increases in net radiation seen in *Mosaic* (9.2 W m^{-2} higher than *Typical* during daytime) and *HUE* (8.6 W m^{-2} higher than *Typical* during daytime). This increase is due to the small fractions of urban spaces that are not captured by the single land cover per grid in the *Typical* model (Fig. 3D), which decreased α (Fig. 2B and C) and ε

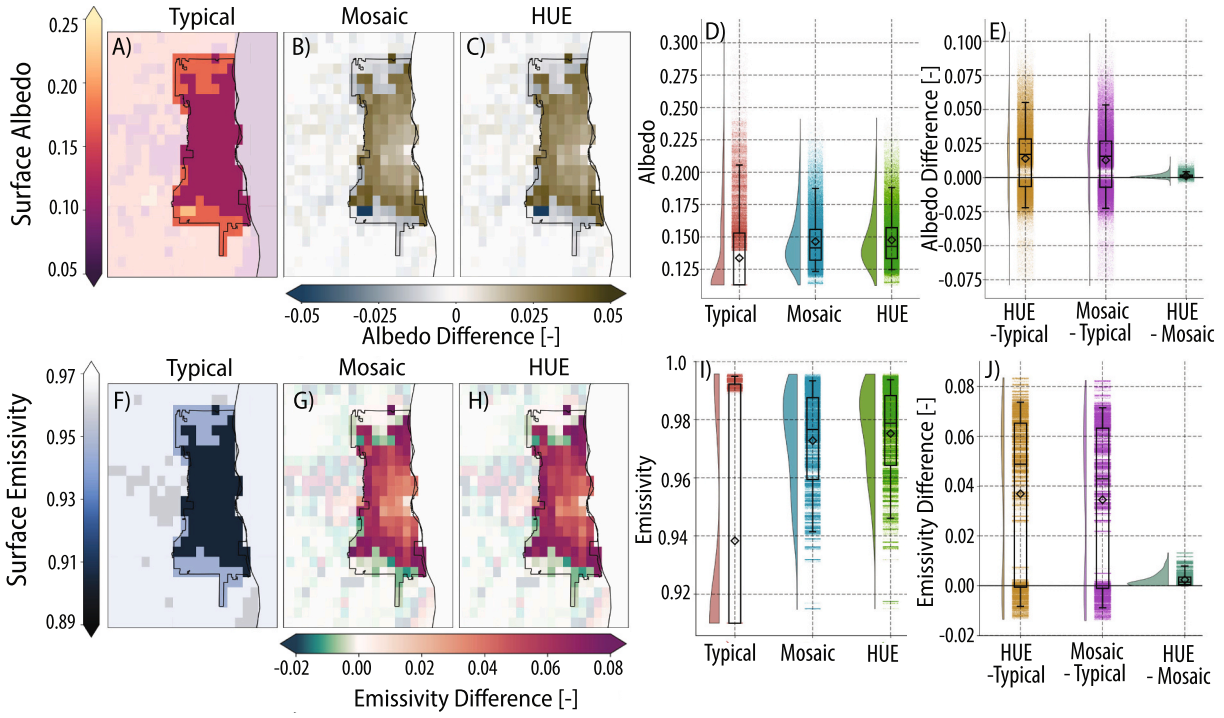


Fig. 2. Albedo and surface emissivity spatial averages over all simulations for *Typical* (A and F), and spatial differences for *Mosaic* (B and G) and *HUE* (C and H) simulations. Full distributions of albedo (D) and emissivity (I) are shown alongside and behind box and whisker plots, with aggregated differences for each time step for *HUE* – *Typical*, *Mosaic* – *Typical*, and *HUE* – *Mosaic* for both albedo (E) and emissivity (J). Whiskers represent the spread of the 5th – 95th percentiles, while the diamonds denote the means of each distribution.

(Fig. 2G and H). Further, decreases in net radiation in areas designated urban are similarly linked to smaller fractions of urban spaces, which tend to increase α and ε. The decreases in suburban regions are larger (*Mosaic – Typical* mean: −11.5 W m^{-2}, *HUE – Typical* mean: −13.1 W m^{-2}) compared to more heavily industrialized locations (*Mosaic – Typical* mean: −5.5 W m^{-2}, *HUE – Typical* mean: −7.2 W m^{-2}; Fig. 3E and F). Differences between *HUE* and *Mosaic* simulations are negligible in non-urban regions, but small differences do occur in suburban (mean: −2.1 W m^{-2}) and heavily urbanized locations (mean: −1.5 W m^{-2}) due to the representation of foliage overlapping pavement in *HUE*.

3.2. *Diverging responses of the partitioning of latent and sensible heat fluxes are driven by available soil moisture*

We broke the 12 WRF simulations into two different categories based on available soil water: a Dry Regime, comprising the July 2012 simulation, and a Wet Regime, which includes Aug 2021, Sep 2016, July 2016, Sep 2019, July 2015, July 2019, Sep 2018, June 2014, July 2017, and June 2018. Introducing vegetation in urban simulations can lead to drying due to interactions between antecedent conditions and enhanced, land cover-specific root water uptake (Alexander et al., 2024; Grimmond and Oke, 1999). This can then lead to differences in energy partitioning at multiple scales (Dou et al., 2023; Santanello Jr. et al., 2019; Seneviratne et al., 2010; Templeton et al., 2018). Comparisons between long-term, uncoupled, land surface simulated soil moisture from May – October over the years 2012–2021 in the urban portion of the MMSD planning region and WRF simulated soil moisture indicates a clear difference between the July 2012 simulation and the other eleven simulations (Fig. 4), which motivated this split for further analysis. The July 2012 simulation includes a notable heatwave that corresponded with a domain-averaged antecedent soil moisture value of 0.205 m^3

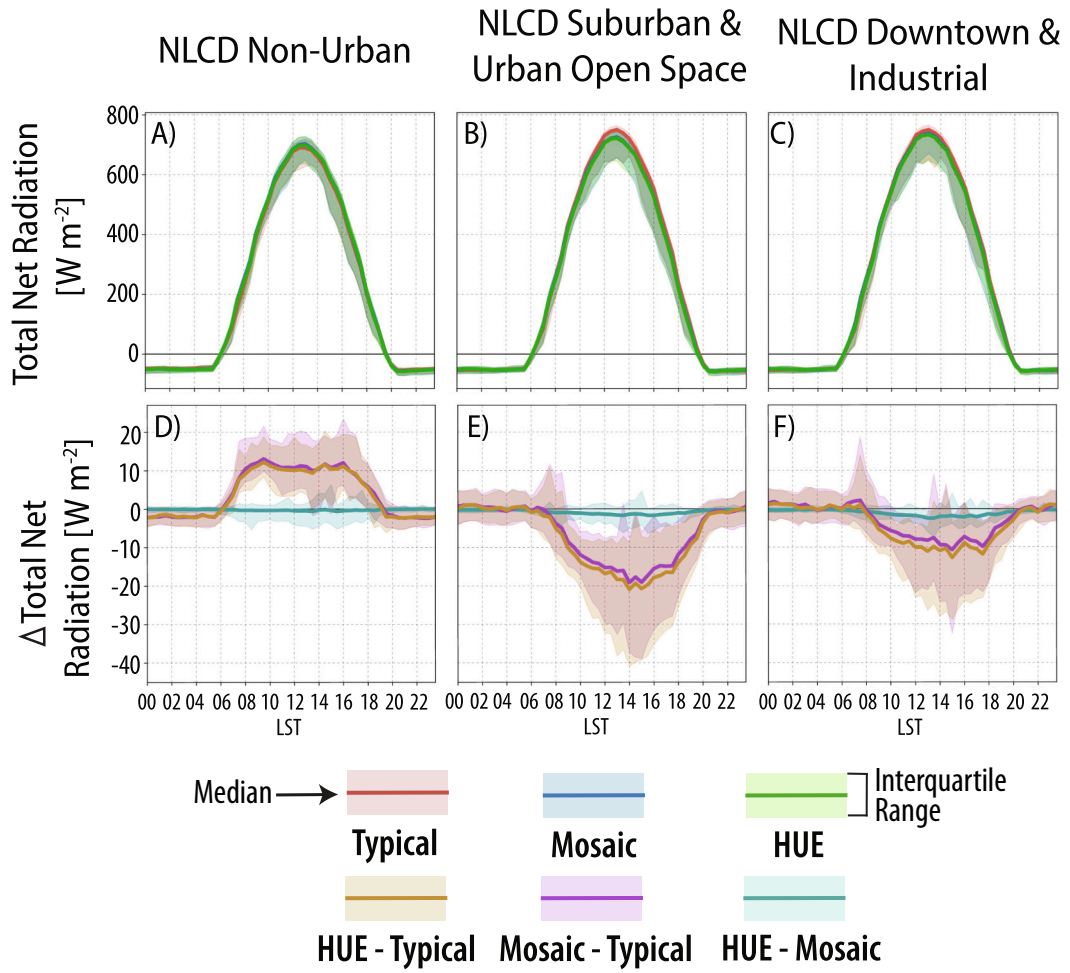


Fig. 3. Average diurnal cycles for total net radiation, with positive indicating radiative flux towards the surface, for *Typical* (red), *Mosaic* (blue), and *HUE* (green) pooled across all simulations. Diurnal cycles are aggregated by dominant land cover types (Non-Urban: A, Open Spaces and Low Intensity: B, and Heavily Urbanized and Industrial: C). Differences in the diurnal cycles of total radiation aggregated *HUE-Typical* (gold), *Mosaic-Typical* (purple), and *HUE-Mosaic* (teal) disaggregated by NLCD land cover type (Non-Urban: D, Open Spaces and Low Intensity: E, and Heavily Urbanized and Industrial: F). In both diurnal cycle and difference of diurnal cycle, solid-colored lines denote the median, while the shading shows the interquartile range. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

m^{-3}, which is lower than 96.8 % of days from offline simulations. Due to limited soil moisture availability in this simulation, evapotranspiration (ET) is reduced by more than 50 % by soil water availability (Fig. 4), while ET is limited in other coupled simulations ranging between 1 % and 15 %.

To introduce the differences that arise between *HUE* and *Mosaic*, we present results for both sensible heat (*H*) and latent heat (*LH*) fluxes for the pixel where the largest shift in land cover occurs in the MMSD planning region (Fig. 5). This pixel, located just east of the city center in a moderate density urban landscape interspersed with mature trees, row homes, and small parks, is composed of 57 % urban, 22 % tree cover, and 20 % turfgrass in *Mosaic* simulations. In *HUE, 16 %* of the pixel is categorized as tree over pavement, and 45 % of the urban land (26 % of the pixel area) routes water to turfgrass when there is rainfall.

In this pixel, differences in *H* and *LH* diverge based on soil water availability, creating diverging effects. When soil moisture is abundant (Wet Regime), *LH* is higher by 27 W m^{-2} on average during the day in *HUE*, resulting in 0.53 mm more of ET per day compared to *Mosaic* simulations (Fig. 5B). *HUE* in this case gives more tree area for ET to occur. Consequently, less energy is partitioned into *H* within *HUE* simulations when there is ample soil water (−41 W m^{-2}) compared to *Mosaic* simulations, to meet atmospheric evaporative demand (Fig. 5 A). When soil moisture is limiting (Dry Regime), *LH* is lower by −31 W m^{-2} in *HUE*, resulting in a reduction in ET of −0.61 mm d^{-1} compared to *Mosaic* (Fig. 5D). Thus, more energy is partitioned into *H* production, especially in the afternoon in *HUE* compared to *Mosaic* (maximum of 91 W m^{-2} at 16 LST; Fig. 5C). Notably, under ample soil water conditions (Wet Regime), *H* profiles are reduced overall by *HUE*, while the soil moisture-limited (Dry Regime) conditions mainly change the average afternoon flux profile (Fig. 5A and C). The changes in *LH* are present throughout the daytime and are enhanced in the Wet Regime or diminished in the Dry Regime (Fig. 5B and D). This effect is tied to the increased ability for *HUE* to produce ET: more ET during times of ample soil water dries soil out quicker, and during times of extremely low soil moisture (Dry Regime), *HUE* has likely already used much of its soil water and thus cannot generate much ET even though it has a larger capacity to do so. While important hydrologically for runoff reduction during rainfall, the shallow rooting turfgrasses are not well adapted helping mitigate heat during the Dry Regime simulation, with an increase of only 3 W m^{-2} for *LH* and decrease of 2 W m^{-2} for *H*.

Compared to *Typical* simulations, peak *H* in non-urban locations are higher in *HUE* and *Mosaic* during the Dry Regime case (*Typical*: 387 W m^{-2}, *Mosaic*: 505 W m^{-2}, *HUE*: 525W m^{-2}) and in the Wet Regime case (*Typical*: 141 W m^{-2}, *Mosaic*: 180 W m^{-2}, *HUE*: 181 W m^{-2}; Fig. 6A), likely due to the increase in urban region representation in *Mosaic* and *HUE* within these peri-urban regions. However, the differences between *HUE* and *Mosaic*-based *H* are strongly influenced by soil moisture availability (Fig. 7A, B, and C). Differences in *H* between *Mosaic* and *HUE* in the Wet Regime are near zero in non-urban spaces, while in the Dry Regime case, *HUE* values are lower on average during the day (*HUE*: 270 W m^{-2} vs. *Mosaic*: 275 W m^{-2}), but generate higher at peak *H* by 40 W m^{-2} (Fig. 7A).

In urban spaces, *Typical* simulations partition almost all available energy to *H* due to the lack of vegetation to promote ET, with peaks converging to nearly the same values in suburban (Wet Regime: 672 W m^{-2}, Dry Regime: 694 W m^{-2}) and heavily urbanized (Wet Regime: 673 W m^{-2}, Dry Regime: 714 W m^{-2}) areas. Similarly, *Mosaic* and *HUE* simulations partition more energy to *H* in suburban and heavily urbanized land covers (Fig. 6B and C). The differences in suburban regions between *HUE* and *Mosaic* (Fig. 7B) indicate that the ample soil water of the Wet Regime leads to a reduction of *H* throughout much of the day (peaking at −18 W m^{-2} at 12 LST), while the limited soil water of the Dry Regime leads to increases in *H* (peaking at +32 W m^{-2} at 14 LST), as well as in its variability. Compared with *Mosaic*, *HUE* in heavily urbanized locations see a stronger decrease of *H* on average (peaking at −30 W m^{-2} at 12 LST) in Wet Regime cases while the Dry Regime case show a decrease *H* in the morning hours (average of −10 W m^{-2}) and slight increases *H* in the afternoon (average of +6 W m^{-2}; Fig. 7C).

LH is larger in the Wet Regime and smaller in the Dry Regime in *HUE* simulations compared to *Mosaic* (Fig. 6D, E, and F).

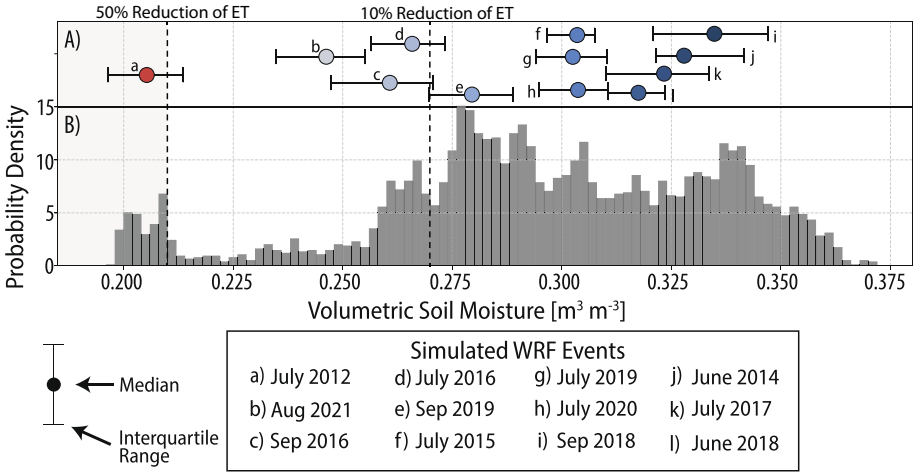


Fig. 4. Urban soil moisture median and interquartile range for all 12 WRF simulations that utilized the *HUE* land cover (A). Coloring of median is based on median soil moisture value. Empirical soil moisture probability density from growing seasons (May – October) from 2012 to 2021 from long-term continuous offline simulations of *HUE* model (B). Vertical dashed lines denote when the regional evapotranspiration is reduced by 10 % and 50 % or more due to limited soil water.

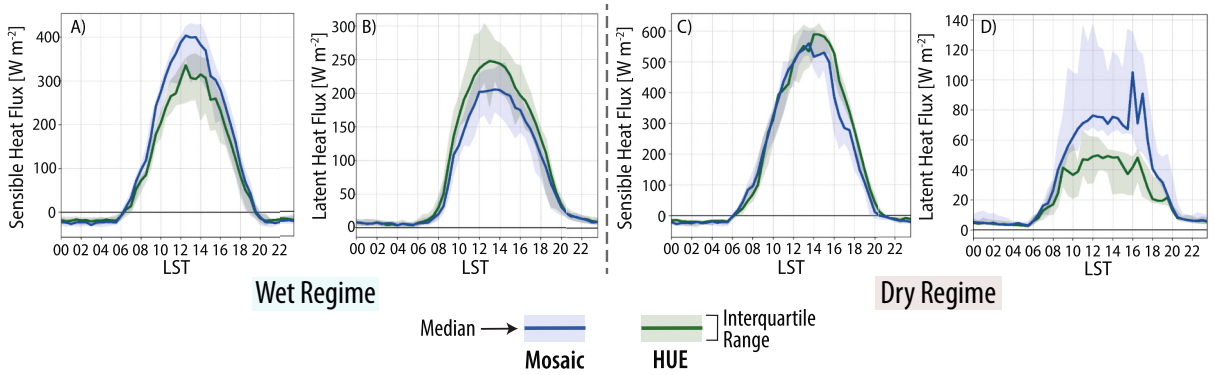


Fig. 5. Diurnal patterns of Sensible Heat Flux (A) and Latent Heat Flux (B) for the pixel with the largest amount of additional tree canopy that overhangs pavement (15.9 %) in *HUE* during the Wet Regime simulations for *Mosaic* and *HUE*. Additionally, Sensible Heat Flux (C) and Latent Heat Flux (D) at the same pixel for Dry Regime Simulations. Solid lines show the median of the variable, while shading shows the interquartile range.

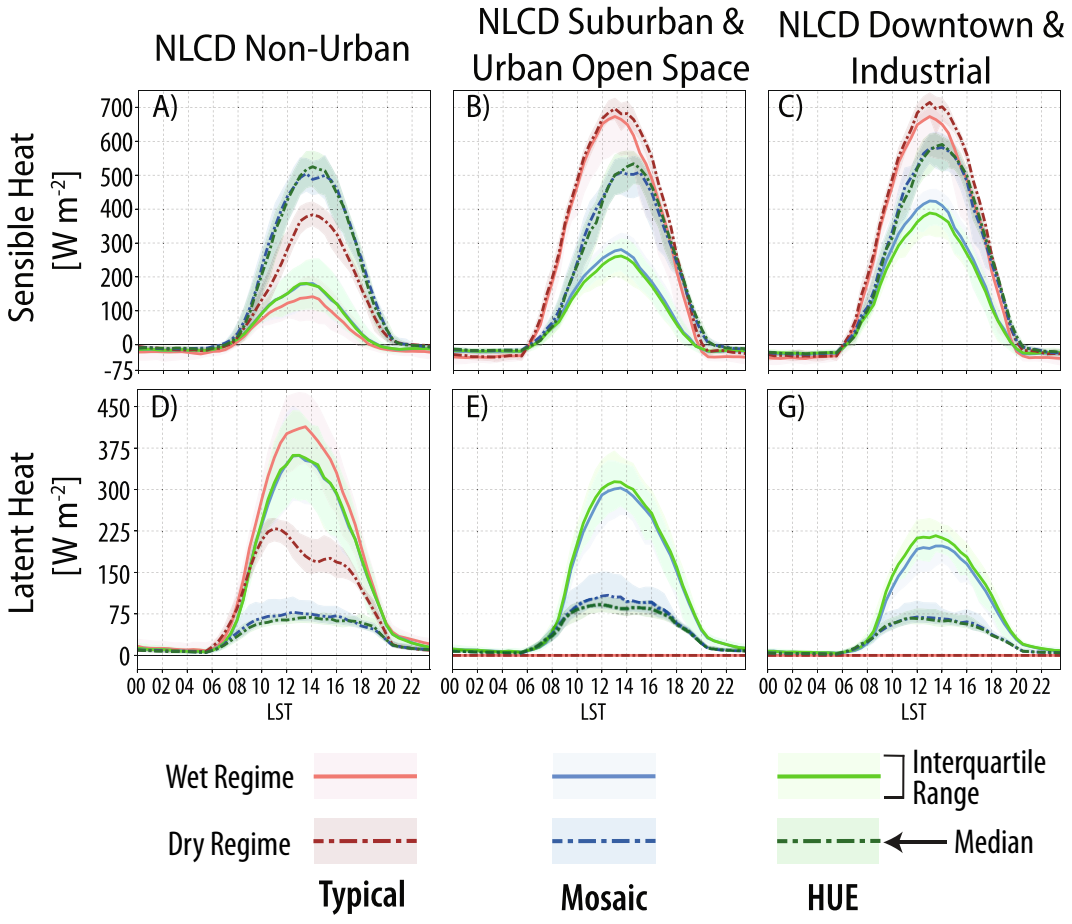


Fig. 6. Diurnal cycles for sensible and latent heat aggregated by NLCD land cover for the Wet Regime (less than 50 % evapotranspiration reduction due to available soil water) and Dry Regime (more than 50 % evapotranspiration reduction due to available soil water). Positive sensible heat indicates heating of the surface by the atmosphere, and positive latent heat indicates evapotranspiration. Solid colored lines demarcate the median of wet regime simulations, dashed lines demarcate the median of the dry regime simulation, and shading shows the interquartile range.

Differences in *LH* between *HUE* and *Mosaic* simulations are marginal in non-urban spaces during Wet Regime cases. Differences in the Dry Regime case average − 6 W m^{-2}, indicating that less energy is used for ET in the *HUE* simulation. In Wet Regimes, *HUE* increases *LH* peaks in suburban areas (*Mosaic*: 300 W m^{-2}, *HUE*: 314 W m^{-2}) and in heavily urbanized areas (*Mosaic* 198 W m^{-2} and *HUE* 216 W

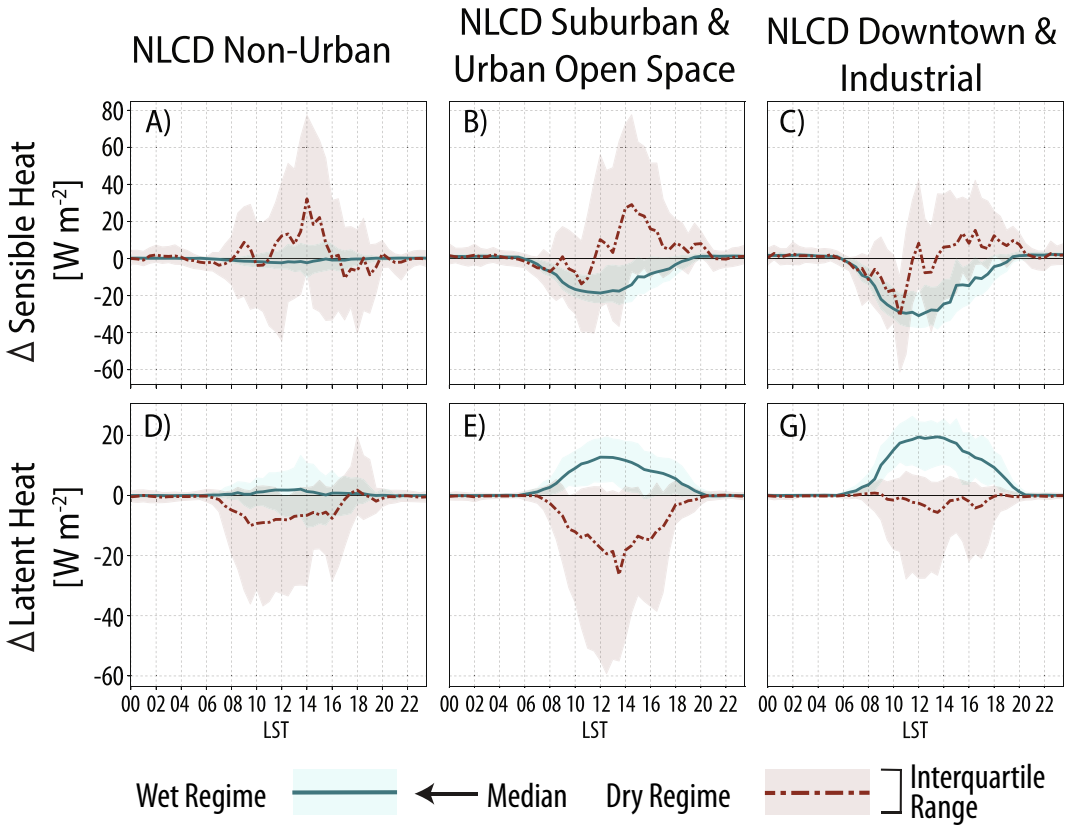


Fig. 7. Differences in diurnal cycles for sensible and latent heat between *HUE* and *Mosaic* simulations aggregated by NLCD land cover for wet regime (blue) and dry regime (red). Positive values indicate more energy partitioned to a given flux in *HUE* simulations. Solid colored lines demarcate the median of wet regime simulations, dashed lines demarcate median of the dry regime simulation, and shading shows the interquartile range. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

m^{-2}). Differences in *LH* between *HUE* and *Mosaic* are larger (peaking at +20 W m^{-2}) in heavily urbanized spaces on average compared to suburban areas (which peak at +12 W m^{-2}) with similar variability. The limited soil water of the Dry Regime case does not follow a traditional diurnal pattern in which *LH* increases till near midday and decreases till night in *HUE* and *Mosaic*. Instead, *LH* increases modestly and then flattens out near ~75 W m^{-2} from 8 to 18 LST in all land covers. In Dry Regime suburban regions, *HUE* changes the amount of *LH* by an average of −10 W m^{-2}, with a wide range of variability (Daytime Average IQR of 32 W m^{-2}). Differences between *HUE* and *Mosaic* is smaller in heavily urbanized locations during the Dry Regime, but still indicate *HUE* partitions less energy to *LH* throughout the day.

3.3. Spatial effects on average daytime near surface air temperature and humidity

Due to the heterogeneity of multiple land cover types, *Mosaic* and *HUE* simulations reduce 2 m air temperatures (*T2*) compared to *Typical* simulations, while *T2* values in *HUE* are cooler compared to *Mosaic* when there is ample soil water (Fig. 8) and warmer when soil moisture is limiting (Fig. 10). Urban average *T2* in *Typical* simulations are 27.2 °*C*, while both *Mosaic* and *HUE* averages 25.2 °*C* for the Wet Regime case, corresponding to an average decrease of 1.9 °*C*. When comparing *HUE* minus *Mosaic* in the Wet Regime case, there is subtle warming in non-urban regions and cooling in heavily urbanized areas (maximum cooling magnitude of 0.18 °*C*) proportional to the amount of tree canopy overlying pavement added (Fig. 8E). Dry Regime simulated *T2* averages are higher than Wet Regime (*Typical*: 30.3 °*C*, *Mosaic*: 28.8 °*C*, *HUE*: 28.9 °*C*), but the decreases between *Typical* and other simulations are similar across all MMSD (*Mosaic* – *Typical*: -1.5 °*C*, *HUE* – *Typical*: -1.4 °*C*). *HUE* and *Mosaic* comparisons for the Dry Regime show much of the western MMSD planning region to be warmer (max warming of 0.4 °*C*), while locations closest to Lake Michigan are cooler (Fig. 10E). Cooler temperatures near Lake Michigan are driven by differences in localized land-lake breeze circulations, rather than differences in land cover and energy partitioning as seen in the rest of the region.

Mosaic and *HUE* simulations increase 2 m specific humidity (*Q2*) compared to *Typical* simulations, while *Q2* in *HUE* is wetter compared to *Mosaic* when there is ample soil water (Fig. 9) and drier when soil moisture is limiting (Fig. 11). Urban region 2 m specific humidity in *Typical* simulations are low (mean: 13.4 g kg^{-1}) due to the lack of vegetation, while *Mosaic* (mean: 14.8 g kg^{-1}) and *HUE* (mean: 14.9 g kg^{-1}) are higher during the Wet Regime case. There is little difference in *Q2* between *HUE* and *Mosaic* when there is

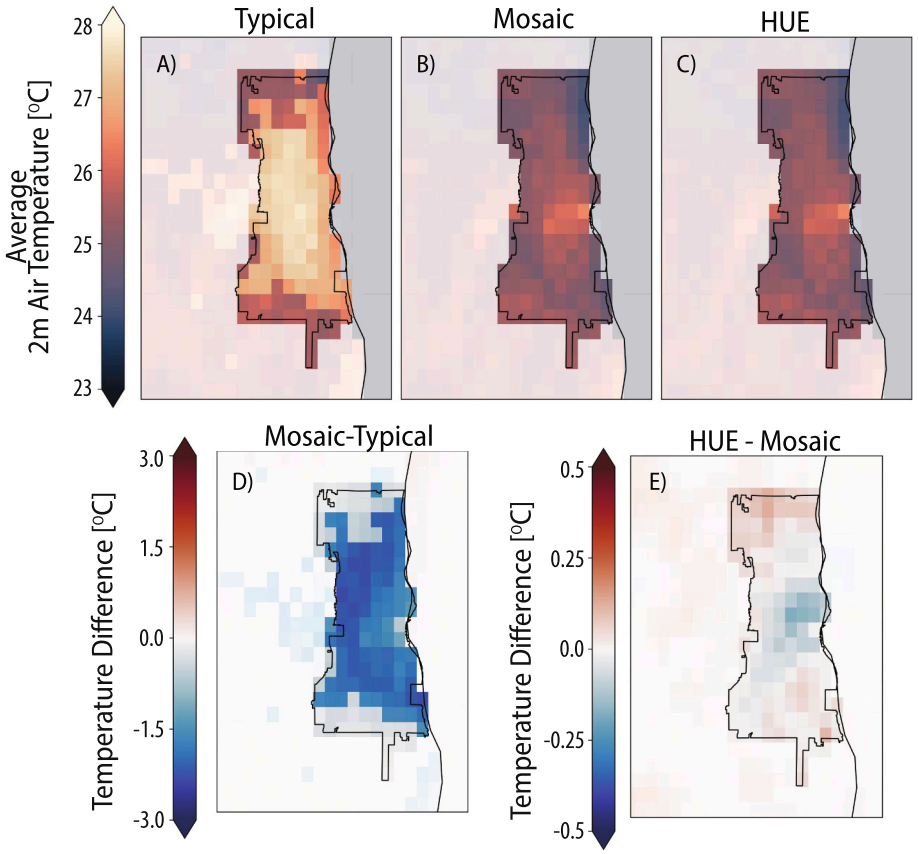


Fig. 8. Average daytime 2 m Air Temperature (*T2)* for *Typical* (A), *Mosaic* (B), and *HUE* (C) for Wet Regime simulations. Temperatures differences between *Mosaic* – *Typical* (D) and *HUE* – *Mosaic* (E), where red indicates a warmer simulation listed first in each panel title. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ample soil water in non-urban and suburban regions, but larger increases (maximum increase of 0.3 g kg^{-1}) in more urbanized locations where larger fractions of tree canopy have been added (Fig. 9E). In the Dry Regime case, *Mosaic* and *HUE* simulations create wetter conditions on average (*Typical*: 11.8 g kg^{-1}, *Mosaic*: 12.8 g kg^{-1}, *HUE*: 12.6 g kg^{-1}). Most of the MMSD planning region is drier in *HUE* compared to *Mosaic* (maximum decrease of −0.5 g kg^{-1}), while locations closest to Lake Michigan are slightly wetter, implying that differences are influenced by proximity to Lake Michigan (Fig. 11E).

3.4. Comparison with observations

Comparison between model results and weather stations located in the urban areas of the MMSD planning region (Fig. 1B) show that *HUE* generally increases simulation skill compared to both *Mosaic* and *Typical* for *T2* (Fig. 12). *Typical* simulations are generally warm-biased, with mean biases (MB) ranging from +1.2 $^{\circ}C$ to +3.3 $^{\circ}C$ depending on the time period (Fig. 12G) and this warm bias is relatively uniform in space (Fig. 12D). Further comparison between *Typical* simulations and observations reveals high correlation (Pearson's R: 0.88), though there is also high RMSE (3.07 $^{\circ}C$) for *T2*. *HUE* and *Mosaic* simulations are cool biased (no more than −1 $^{\circ}C$) in three events and warm biased in the remaining nine (no more than 1.25 $^{\circ}C$; Fig. 12G). Both *HUE* and *Mosaic* reduce bias at all stations compared to *Typical* (Fig. 12E and F). This increase in skill for *HUE* and *Mosaic*, compared to *Typical*, is also reflected in lower RMSE values, particularly in the 10th and 90th quantiles (Fig. 12A, B, and C). Despite their similarities, *HUE* simulations generally match observations more closely than *Mosaic*, with a higher Pearson's R (*HUE*: 0.88 *Mosaic*: 0.87) and lower RMSEs, though differences are small.

Weather station comparisons of *Q2* show that *HUE* and *Mosaic* perform better than *Typical* simulations (Fig. 13). *Typical* simulations are consistently dry biased (Fig. 13A, D, and G). *HUE* and *Mosaic* perform similarly in space, with three comparisons with weather stations exhibiting a wet bias, the largest of which is the Milwaukee Airport (Fig. 13E and F). Both *HUE* and *Mosaic* reduce MB throughout all events, ranging generally between ±1 g kg^{-1} (Fig. 13G), and are more correlated with lower RMSE relative to the *Typical* simulation (Fig. 13B and C). In contrast to the *T2* comparison results, *HUE* is slightly less correlated with observations than *Mosaic* (*HUE*: 0.77, *Mosaic*: 0.76), though *HUE* has lower MB in eight of the twelve WRF simulations (Fig. 13G).

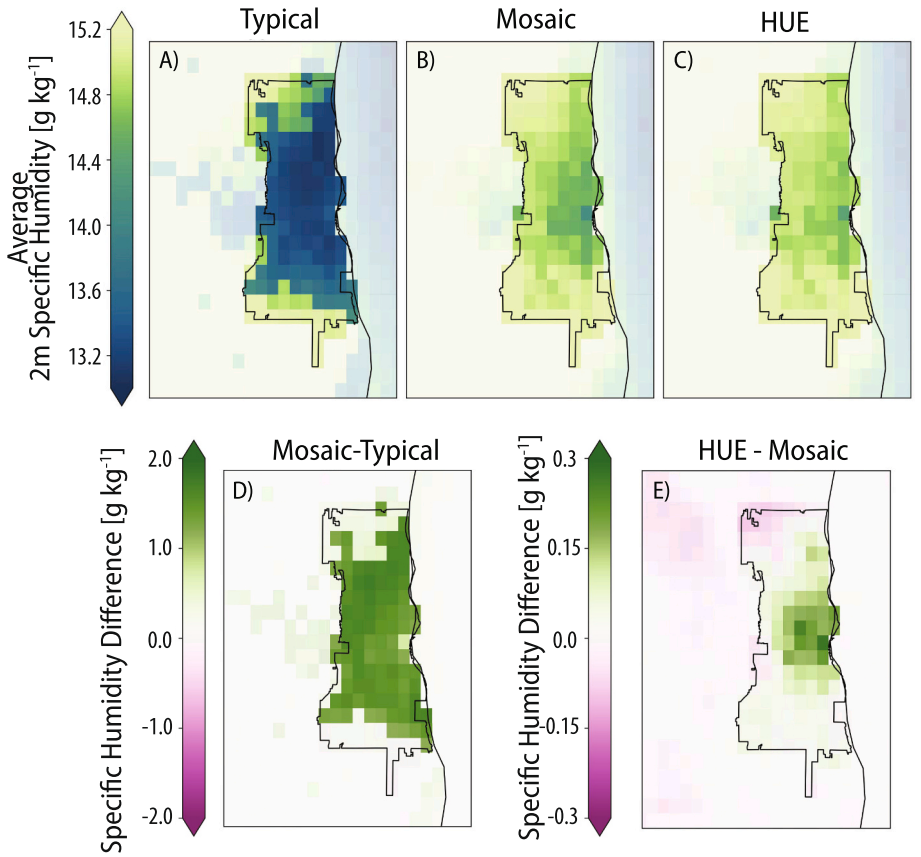


Fig. 9. Average daytime 2 m Specific Humidity (*Q2*) for *Typical* (A), *Mosaic* (B), and *HUE* (C) for Wet Regime simulations. Humidity differences between *Mosaic* – *Typical* (D) and *HUE* – *Mosaic* (E), where green indicates a wetter simulation listed first in each panel title. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

4.1. Implications for coupled land-atmosphere modeling under current urban conditions

The bulk model (*Typical*) is less skillful at simulating *T2* and *Q2* due to the lack of hydrologic and vegetation representation in urban regions. Adding sub-grid representation of urban heterogeneity through multiple land cover types (*Mosaic* and *HUE*) increases the skill of both *T2* and *Q2* simulations (Fig. 12 and Fig. 13), aligning with other studies that have found improvements in near surface temperature and humidity in coupled land-atmospheric models in urban regions with mosaicking schemes (e.g., Jänicke et al., 2017; Li et al., 2018; Sharma et al., 2016b; Wang et al., 2022; Zhang et al., 2021). When using *Noah-MP HUE*, *T2* simulations are further improved, with lower RMSE and higher correlation compared to the *Mosaic* simulations (Fig. 12B and C). Simultaneously, *Noah-MP HUE* is comparable to *Mosaic* when simulating *Q2,* with lower bias in most of cases examined (Fig. 13G) but slight degradation in RMSE and correlation (Fig. 13B and C). Skillful simulation of *T2* and *Q2* are critical for prediction and impacts of thermal stress on health (Hoffman et al., 2020; Hsu et al., 2021; Minard, 1957; Tong et al., 2021). Improved near-surface air temperature and humidity also provides better representations of local scale processes that lead to convective rainfall (Sharma et al., 2016b, 2021; Shepherd, 2005; Yang et al., 2024), which continues to be critical for hydrologic and hydraulic planning in urban areas (Ivanov et al., 2021; Tran et al., 2024; Wright et al., 2019). Simulation skill could be further improved by incorporating vegetation shading and water transfer from urban surfaces to vegetated soils within more advanced urban models, like the urban canyon (Kusaka et al., 2001; Kusaka and Kimura, 2004) or the Building Effect Parameterization (Martilli et al., 2002b). However, our study aimed to isolate differences caused by land cover representation and new, sub-grid, hydrologic processes in urban areas, which would not be possible if model formulation also changed.

The main driver of differences in urban surface air temperature and humidity is tied not to increases or decreases in available energy, but instead to differences in partitioning of net radiation into *H* and *LH*. The presence of urban vegetation improves simulation skill due to enhanced ET (comparing *Mosaic* and *HUE* to *Typical*). Further, hydrologic process integration adds a new layer of sensitivity to flux partitioning based on available soil water (comparing *Mosaic* to *HUE*). In Wet Regime simulations, *HUE* was cooler and wetter compared to *Mosaic* and *Typical* (Fig. 8 andFigure 9) due to ample soil water to enhance ET via tree cover overhanging pavement, and

to a lesser extent increased turfgrass transpiration from rainfall runon after storms. This aligns with a recent study that incorporated tree cover into the Urban Canyon Model within WRF that shows evaporative cooling in ample soil water conditions (Ryu et al., 2025). The Dry Regime simulation generally led to warmer and drier conditions in *HUE* simulations compared to *Mosaic* (Fig. 10 and Fig. 11) which was due to the decreased soil water availability in *HUE* itself. A consequence of the more realistic treatment of tree cover in *HUE* is the proclivity for soils to dry faster, as the increased amount of tree cover can dry soils out quicker than traditional mosaicking schemes (see Fig. 5 in Alexander et al., 2024). This results in cooling Nevertheless, the improved *T2* and comparable *Q2* error metrics in *HUE* relative to *Mosaic* (Fig. 12 and Fig. 13) suggest that *HUE* simulations may hew more closely to reality due to this higher sensitivity. While this finding may seem contrary to intuition of role of trees in modulating temperature in urban spaces for the Dry Regime, it appears to highlight an important and overlooked process. Specifically, drying out of soil in the *HUE* simulations due to increased use of soil water from vegetation (e.g. tree canopy that over hangs pavements) causes increases in *H* and decreases in *LH*, especially in suburban areas (Fig. 7B and Fig. 7E). Despite drastically different hydrometeorological conditions, *H* diurnal profiles are similar in the Wet and Dry Regimes for *Typical* simulations (Fig. 6). Thi.

Our results align with other studies: measurements from both eddy covariance and scintillometry show that urban spaces predominantly partition net radiation to *H*, while *LH* fluxes are highly variable depending on the time of year and the amount of vegetation present (Christen and Vogt, 2004; Dou et al., 2023; Ferreira et al., 2024; Ward et al., 2013, 2015). Goldbach and Kuttler (2013) highlight that enhancement of *H* or *LH* fluxes are directly tied to soil water availability, even in urban areas; higher sensitivity has been reported in heavily urbanized regions than suburban areas. Both our *Mosaic* and *HUE* simulations support this. These simulations preferentially partition energy to *LH* flux in suburban land covers (Fig. 6G), creating a Bowen Ratio, i.e. the ratio of sensible to latent heat fluxes (Oke et al., 2017; Stull, 1988), of ~0.8, and simulating a Bowen Ratio of ~2 in heavily urbanized regions. These values are representative of urban spaces worldwide, as summarized in Table 1 of Goldbach and Kuttler (2013).

4.2. Implications for urban greening plans

Wet Regime simulations of *Mosaic* and *HUE* align with the conventional understanding of the effects of urban greening: additional vegetation enhances evaporative cooling (Aram et al., 2019; Balany et al., 2020; Gunawardena et al., 2017) and results in a net reduction in air temperature and an increase in urban humidity (Fallmann et al., 2016; Papangelis et al., 2012; Schwaab et al., 2021; Sharma et al., 2016a; Wang et al., 2023b; Zhou et al., 2017). Differences between *HUE* and *Mosaic* in air temperature are relatively small, ranging between 0.1 and 0.4C cooler on average (Fig. 8). Differences between *Mosaic* and *HUE* are driven by the amount of

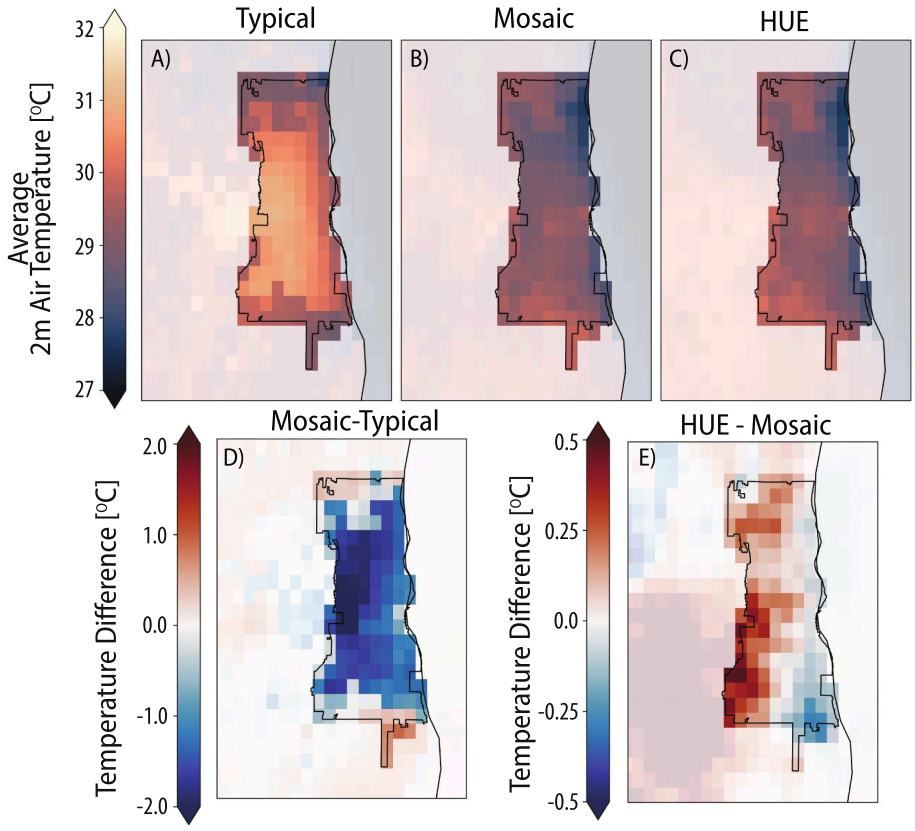


Fig. 10. As in Fig. 8, but for the Dry Regime Simulation.

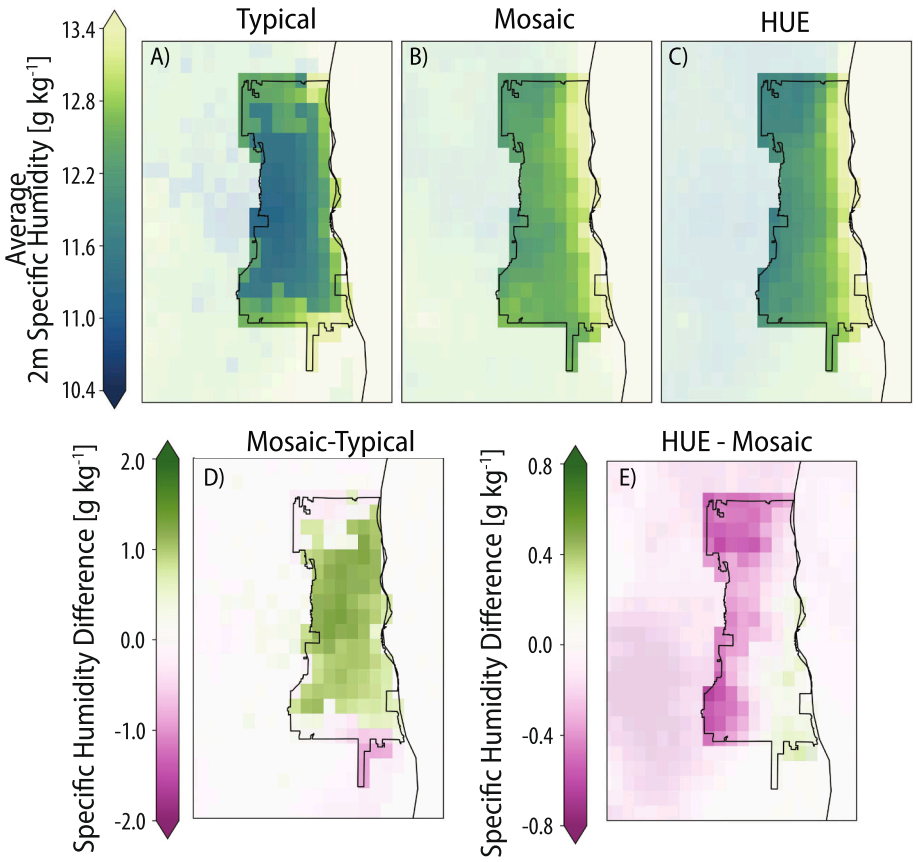


Fig. 11. As in Fig. 9 but for the Dry Regime Simulation.

otherwise neglected tree cover that is 'added back' to urban spaces in the latter simulations: urban spaces with a tree canopy that overlaps sidewalks, driveways, roadways, etc. are classified as urban in *Mosaic* but not in *HUE,* resulting in more overall tree canopy in the latter simulation. Temperature differences in this study between *HUE* and *Mosaic* are within the observed cooling of urban trees outside of parks across 29 studies summarized in Fig. 13 of Knight et al. (2021), indicating that adding tree cover over pavement in *HUE* is generally aligned with observed trends. Further, while this seemingly small difference in temperature and humidity may seem inconsequential, changes on the order of 0.4C can result in clear reductions of heat related mortality (Iungman et al., 2023).

Throughout the MMSD region, *HUE* adds on average 4 % more tree canopy (which is over pavement) and causes cooler and moister conditions. While this is representative of current conditions, the MMSD area is rapidly changing, with an aggressive greening plan that aims to capture the first inch of rainfall throughout the region by widespread implementation of urban greening like tree canopy, green roofs, permeable pavements, and downspout disconnection (Milwaukee Metropolitan Sewerage District, 2013) Specifically, the MMSD greening initiative includes adding ~50 km^2 of tree canopy (Milwaukee Metropolitan Sewerage District, 2013), much of which will overhang pavements and enhance the cooling benefits. Thus, *HUE* can be a valuable tool to help evaluate the impacts on surface air temperature and humidity due to complex feedbacks in shading, water stress, and ET which cannot be captured using just a mosaicking scheme.

The Dry Regime simulation indicates that vegetation may actually intensify heatwaves in urban regions if there is insufficient soil moisture to support ET. Although the Dry Regime consists of a single event, and thus could be an outlier in the temperate climate studied here, these results could be an indication of how expansion of tree cover could make (sub)urban areas drier and warmer in water limited regions. Indeed, a small number of both observational (Grimmond et al., 1996) and modeling (Schlaerth et al., 2023) studies conducted in semi-arid southern California indicate that greening can lower urban albedo and lead to increased air temperatures. A simple way to counteract this warming could be to route runoff from impervious surfaces to these trees in the street terrace, as we have done for lawns (but not trees) in *HUE*. Alternatively, the use of urban irrigation practices could help to reduce plant stress and promote ET, which has been shown at the local scale to increase thermal comfort (e.g. reduce air temperatures due to enhanced ET and direct soil evaporation; Gobatti et al., 2025). This solution poses problems, however, in locations with already scarce water resources, at least without additional actions like rain harvesting or watering with non-potable water (Cheung et al., 2024; Livesley et al., 2021; Zhang et al., 2023).

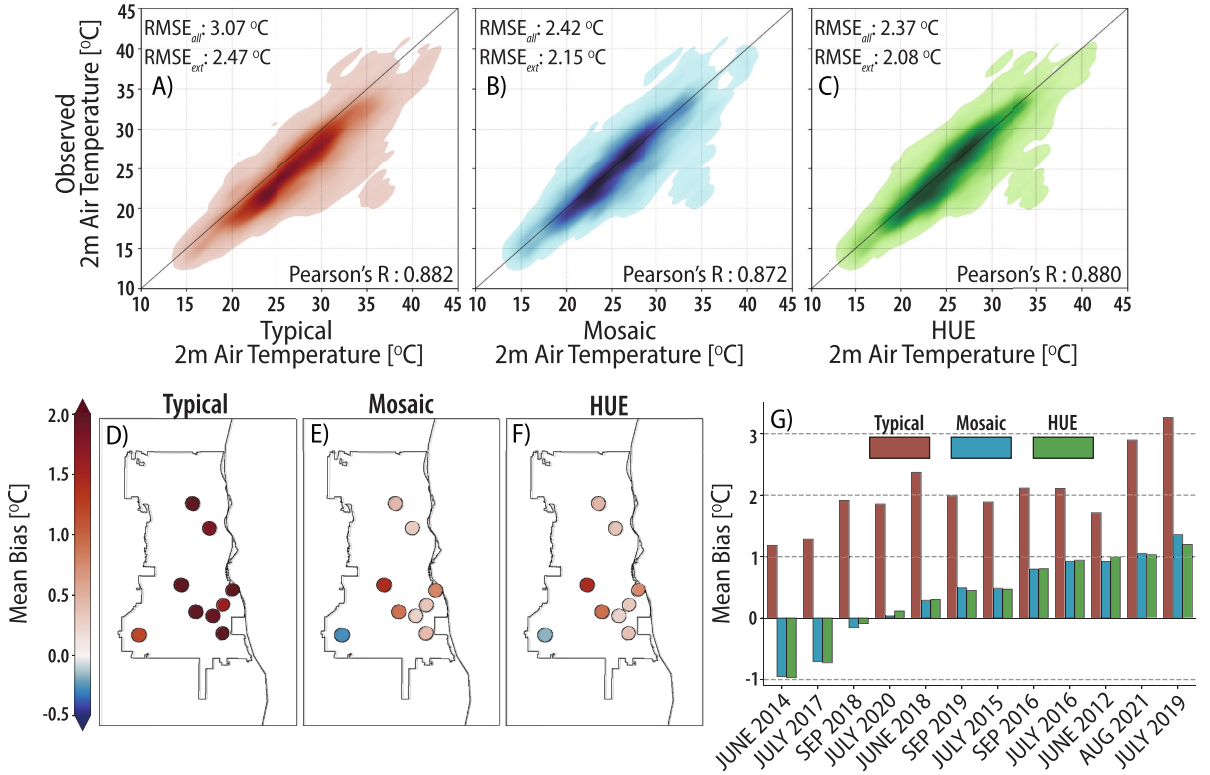


Fig. 12. Comparison of simulated and observed hourly 2 m Air Temperature *Typical* (A), *Mosaic* (B), and *HUE* (C). Root mean square error for the entire set of comparisons ($RMSE_{all}$), for times when observations are above (below) the 90th (10th) quantile ($RMSE_{ext}$), and Pearson's Correlation Coefficient (R) for comparison. Maps show spatial distribution of mean bias for *Typical* (D), *Mosaic* (E), and *HUE* (F) at weather stations across the Milwaukee region across all events. Bar chart (G) shows the average mean bias for each simulation event across all stations.

4.3. Limitations of this study and future research avenues

A critical aspect that could be affecting our results 1) non-characterization of critical urban hydrologic practices and 2) the underlying distribution of land cover derived for this study. The Dry Regime simulation points out that neglecting the runoff routing to other vegetation, not just turfgrass, could be reducing ET and increasing urban heat due. While this is a process that is present within cities, our intent was to investigate practices that were widespread in the Milwaukee region, which is downspout disconnection due to city ordinances and policy. Nevertheless, we intend to add this hydrologic feedback into our simulations, though more work is needed on to carefully identify trees and pavements that are co-located such that that they could be hydrologically connected to pavements via runoff routing. Identification of these co-located features would likely need to use a more sophisticated image detection like that used for tree canopy already in this study (Bosch, 2020; Yang et al., 2009). Further, our use of open-source data to derive land cover through GIS layers and NAIP imagery is similar to the methods used to generate the NLCD thematic datasets (Dewitz, 2021), but are applied at a localized scale. Some land cover types, mainly tree overlapping pavement and turfgrass, are not categorized in contemporary land cover datasets, but are accounted for in thematic data for canopy cover or impervious surfaces (Greenfield et al., 2009). We suspect that our datasets have similar errors in impervious surface estimates, average underrepresentation (e.g., missing impervious surfaces) of 2 % (Wickham et al., 2020) and tree canopy, average of 5 % overrepresentation (Pourpeikari Heris et al., 2022). To ensure the functionality of *HUE* in other urban regions, a more robust method using imagery data to classify land cover in complex urban environments could be a future area of research.

This study considered only a limited number of hydrometeorological conditions in a single, mid-sized coastal city in a cool continental climate. Results are likely to differ based on location due to the influences of urban surface roughness, background climate, and meteorology on urban heat and humidity (Li et al., 2019; Manoli et al., 2019; Zhao et al., 2014). Furthermore, our study does not investigate long term (e.g. seasonal or annual) changes that may occur through the integration of urban hydrology and heterogeneity. Preferential partitioning of rainfall to soil moisture storage, seen in offline tests of *Noah-MP HUE* (i.e., not coupled with WRF; Alexander et al., 2024) are likely to generate larger differences in near surface atmospheric variables (e.g. wetter day-to-day conditions and more evaporative cooling), and may even lead to differences in the local climate as a whole. To investigate potential effects, multiyear simulations at multiple locations are needed to draw firmer conclusions on the effectiveness of urban greening as a whole. We intend to perform multiple season-long simulations to explore the effects of widespread greening initiatives within the MMSD region based on

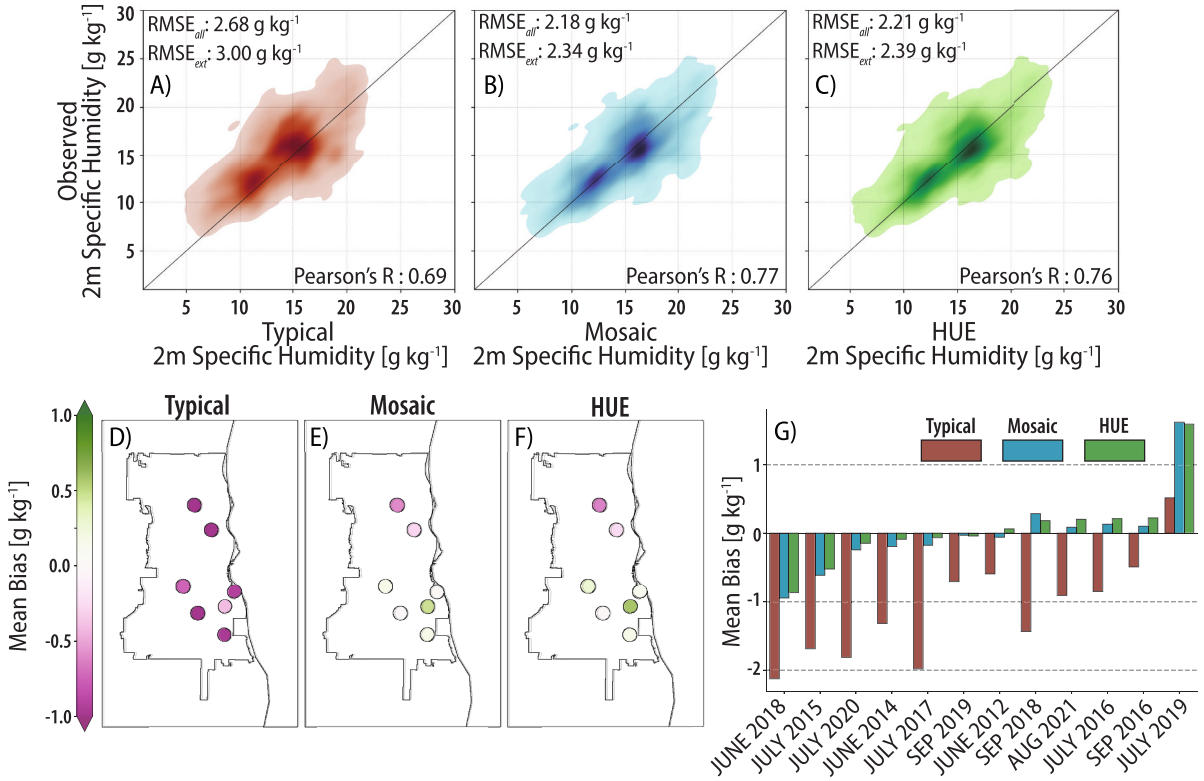


Fig. 13. As in Fig. 12, but for 2 m Specific Humidity.

their current plans (Milwaukee Metropolitan Sewerage District, 2013). This will not only explore different conditions like warm years and wet years, but also different aspects of the greening initiatives such as the roles of large scale depaving, green roof implementation, and the aforementioned 50km^2 expansion of tree cover in climate mitigation effects.

5. Conclusions

We explored the impact of urban heterogeneity and small-scale (< 3 km × 3 km) urban hydrology on city-scale land-atmospheric simulations, focusing on how these factors affect energy flux partitioning and near-surface temperature and specific humidity across a range of 12 hydrometeorological events. We compared a single dominant land cover simulation (Typical Noah-MP), a mosaicking approach (Noah-MP Mosaic), and a newly developed land surface model that integrates urban vegetation and hydrologic processes (Noah-MP HUE) in a temperate, midsized city in the upper Midwestern United States. Our results show that changing Typical Noah-MP to either Noah-MP Mosaic or Noah-MP HUE increases the albedo and emissivity, resulting in a net decrease in available radiation. Furthermore, HUE has a small decrease in available radiation relative to Mosaic.

When there is ample soil water, urban ecohydrology (*HUE* simulations) within coupled land-atmospheric simulations enhances latent heat fluxes at the expense of sensible heat fluxes relative to a mosaicking approach. This generally leads to wetter and cooler conditions in heavily urbanized locations despite small fractions of urban trees (4 % on average) that overhang pavement or downspout disconnection. When soil water is limited, the opposite occurs: sensible heat in the ecohydrology-enabled simulations is enhanced relative to mosaicking, leading to drier and warmer conditions. HUE simulations improve performance over dominant land cover and mosaic simulations compared to 2 m air temperature and specific humidity observations in our study region, indicating that small scale urban hydrology could be potentially important for refining numerical predictions in cities. We conclude that Noah-MP HUE enables the assessment of urban greening at the city-scale, allowing for the systematic investigation of both recognized benefits—such as runoff reduction and enhanced ET—and their potential effects on local and regional climate.

CRediT authorship contribution statement

G. Aaron Alexander: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Daniel B. Wright:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Carolyn B. Voter:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Steven P. Loheide:** Writing –

review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

G. Aaron Alexander reports equipment, drugs, or supplies was provided by National Center for Atmospheric Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.uclim.2025.102596.

Data availability

Data will be made available on request.

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